When Is Control-Aware Communication Locally Decidable? Information-Structure Limits in Distributed Formation Control

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***Abstract*—Control-aware communication assigns network resources according to the closed-loop consequence of an update. This premise hides an information question: does the transmitting agent possess enough information to determine that consequence? We study finite-dimensional sampled linear systems for which a communication action adds a fixed response to a finite-horizon performance output. The conditional fixed-response quadratic action benefit is an affine functional of an augmented closed-loop state. Building on functional observability, we characterize exact value identifiability from current and finite-history agent information by row-space conditions. We then quantify the irreducible value-estimation radius and the strictly positive local minimax communication-decision regret when compatible values straddle zero. The minimum number of instantaneous noiseless linear scalar statistics needed for several action values equals the rank of their unobservable coefficient projection. Coalition ownership of those statistics and their communication acquisition cost are treated separately.**

The framework is instantiated on implementation-derived sampled formation controllers with delayed and lossy communication. A preregistered registry spanning 72 combinations of 5, 7, and 9 agents, three connected graphs, two pinning patterns, two horizons, and two delays yields 1320/1320 structurally nonidentifiable action functionals. In the frozen five-UAV instance, all ten actual controller actions additionally admit full, unsaturated, dynamically reachable history pairs with identical allowed sender observations and action responses but opposite exact action-value signs. Missing-information compression occurs across leader actions, whereas tested follower action sets show no compression; coalition size varies with topology. Frozen development experiments further show why acquisition matters: centralized exact value has scheduling headroom in a formation-switching regime, whereas charging an ACK-like receiver-information stream can remove that headroom under the packet-frequency metric. The results delimit exact local value-based communication; they do not rule out prior-specific Bayesian decisions or nonlinear policies.

***Index Terms*—Networked control systems, distributed control, information structures, functional observability, value of information, event-triggered control, multi-agent formation.**

I. Introduction

Communication policies for networked control are increasingly asked to send the packet that matters most to the control objective, rather than simply the oldest packet or the packet with the largest state error. Event-triggered control formalizes state- or error-dependent transmissions [1]–[3]; age of information (AoI) formalizes freshness [4], [5]; and value-of-information (VoI) formulations attach a control consequence to an update [6]. These approaches answer different questions. AoI describes how stale a record is, while a finite-horizon control value asks how the closed loop will change if that record is refreshed.

The latter question is meaningful only relative to an information structure. In a distributed formation, the sender knows its own physical and controller state and its transmission history. It need not know the packet currently held by a receiver, the receiver's other stale memories, or remote states that shape the no-action closed-loop trajectory. Acknowledgments can reduce one part of this uncertainty, and ACK-free packet-memory beliefs can be calibrated under a known channel law. Neither fact implies that the total control-action value is locally computable: the cross term between the action response and the global no-action response may remain hidden.

This paper therefore asks a question prior to scheduler design:

> Under what information structures is the conditional fixed-response, finite-horizon benefit of a communication action identifiable, and what additional information must be acquired when it is not?

The question connects decentralized information structures [7]–[9] with functional observability [10], [11]. We do not claim the row-space test itself as new. Rather, we expose the exact communication-action value as the target functional, show that inability to observe its magnitude can extend to its sign, separate the dimension of the missing statistic from the network cost of constructing it, and close these statements on an implemented formation-control model.

The investigation was driven by falsification. A deterministic communication-error bound was analytically valid but produced a Pareto frontier dominated by periodic communication. An online centralized current-state oracle, using the exact action cross term but no future realizations, exposed substantial scheduling headroom in some nonstationary regimes and none in a congestion boundary. Explicit receiver feedback often consumed that headroom. Finally, an exact ACK-free receiver-memory belief was well calibrated but could not identify the baseline component of the action cross term.

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These findings motivated an information-limit result rather than another trigger.

The contributions are:

1) We derive the exact finite-horizon marginal communication-action value as an affine functional of an augmented state for a fixed action response.
2) Using functional-observability geometry, we quantify the irreducible value-information radius and derive deterministic and randomized local minimax communication-decision regret when compatible exact values straddle zero.
3) We prove that the minimum dimension of extra instantaneous linear information for a family of actions is the rank of its hidden projection, and apply it sender-wise to expose shared versus independent missing directions. We separately characterize coalition ownership because statistic dimension is not packet count or distributed acquisition cost.
4) For the sampled formation architecture, a new 72-cell registry across $N = 5, 7, 9$, topology, pinning, horizon, and delay yields 1320/1320 structural nonidentifiability results stable over the declared tolerance grid. In the frozen N=5 instance, we construct and independently replay actual-action opposite-sign witnesses for all ten controller-relevant actions, quantify their heterogeneous practical scale, and retain the ACK-like break-even diagnostic as acquisition-cost sensitivity only.

The results are scoped to exact affine action-value identifiability in the stated sampled linear model. They do not assert that distributed agents can never make useful stochastic decisions, that feedback is always uneconomic, or that adaptive communication is universally inferior to periodic service.

II. Related Work and Positioning

A. From event conditions to stale receiver memories

Event-triggered control replaces periodic sampling by state-dependent conditions while retaining stability or performance guarantees [1]–[3], [12], [13]. Loss and delay make the sender and receiver disagree about the control information in use. Decentralized double-integrator consensus under unreliable links is treated in [14]; ACK and no-ACK event-triggered designs under packet loss appear in [15]; and model-based triggering without ACK appears in [16]. Most closely, Viel et al. maintain packet-loss hypotheses for neighbor-held states in distributed formation control [17]. Thus no-ACK receiver memory inference is established prior art. Our issue is what remains hidden after that receiver-memory uncertainty is represented exactly.

B. Freshness and control value

AoI measures time since generation of the freshest received update [4]. Goal-oriented variants such as age of incorrect information refine freshness by state relevance [5], [18]; a recent multi-agent overview organizes goal-oriented communication by the task information being exchanged [19]. These metrics are useful network summaries but are not generally sufficient statistics for the finite-horizon marginal value of an action. VoI-based feedback control makes that marginal value explicit and can yield optimal threshold structures under its prescribed information model [6]. Recent finite-horizon scheduling under packet drops likewise uses ACK-derived error covariance as a fully observed Markov state [20]. Formation scheduling has also combined value proxies, graph relevance, and resource allocation [21]. We neither introduce the VoI concept nor optimize a scheduler. Bayesian and VoI policies may act optimally under their stated prior and information model without observing a realized clairvoyant value. We instead test the narrower prerequisite for reproducing the full-information, fixed-response value or its sign at a decision.

C. Information structures and functional observability

Who knows what and when is central to decentralized control [7], [8]. Common-information methods can convert specific partial-history-sharing problems into coordinated stochastic control problems [9], but they require a declared probability model and information sharing architecture. Multi-agent return-gap communication clusters local observations using joint action-value vectors and bounds expected loss relative to full observability [22]. That learned stochastic-code problem differs from exact affine reconstruction, local set-membership regret, and instantaneous linear-statistic dimension. Functional observability asks whether selected linear functionals, rather than the full state, can be reconstructed [10]. Its row-space characterizations and target-estimation questions are established, including for large networks [11]. Recent work provides modal and structural characterizations and optimal sensor-placement results [23], sample-based conditions for intermittent or irregular measurements [24], and reduced-order functional-observer existence and pole-placement methods [25]. Structural refinements cover generically diagonalizable systems [26] and fixed-observable-set characterizations [27]; target control/observation duality is developed in [28].

These results imply three explicit boundaries. Our current- and finite-history row-space conditions are applications of functional and sample-based functional observability, not new generic tests. Our coalition search is an instantiated functional sensor-selection problem, not a new optimization class. Finally, the instantaneous missing-statistic dimension derived here is neither minimum sensor count nor functional-observer order. Our current-state test is specifically an instantaneous functional reconstructibility application; it is not a functional-observer existence, detectability, order, or pole-placement result. Our addition is the communication-decision consequence: a conditional fixed-response action-value functional, its local minimax estimation and binary-decision regret, and the separation between simultaneous-action statistic dimension and network acquisition cost.

D. Control–communication co-design

Data-rate theorems and networked-control studies quantify how communication constrains stability and performance [29]–[31]. Modern surveys emphasize that information has several operational meanings in estimation and control [32]. Feedback itself consumes network opportunities and can reverse freshness conclusions [33]. Our break-even analysis is deliberately modest: it asks whether the measured performance headroom can pay for the messages needed to expose a hidden value statistic under a frozen packet-frequency accounting model.

III. Sampled Distributed Formation and Communication Model

A. Formation dynamics

Agent 1 is an analytical leader and $\mathcal{F} = \{2, \ldots, N\}$ are followers. At sample k, follower i has position $p_i \in \mathbb{R}^3$, velocity $v_i \in \mathbb{R}^3$, and input u_i. The implemented semi-implicit double-integrator update is

$$v_i(k+1) = v_i(k) + hu_i(k), \tag{1}$$

$$p_i(k+1) = p_i(k) + hv_i(k) + h^2 u_i(k). \tag{2}$$

For desired offset r_i, define

$$e_i = p_i - p_1 - r_i, \qquad w_i = v_i - v_1. \tag{3}$$

The directed convention is $A_{ij} = 1$ when controller i uses a packet sent by j. The evaluated graph is fixed and its follower-grounded dynamics are Schur under the declared gains. To make the implementation instance explicit,

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}, \quad \pi = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \end{bmatrix}^\top, \tag{4}$$

with $s_i = 1$, $K_p = 1.8$, $K_v = 2.2$, $K_{pL} = 1.5$, and $K_{vL} = 1.8$.

Receiver i stores the newest accepted ordinary neighbor payload $\hat{x}_{ij} = (\hat{p}_{ij}, \hat{v}_{ij})$. Pinned followers separately store $(\hat{p}^L_{i1}, \hat{v}^L_{i1}, \hat{a}^L_{i1})$. Newer-generation acceptance prevents an obsolete reordered packet from replacing newer memory. DATA may be lost or delayed; an ACK architecture may separately report acceptance, but the information-limit theorem only uses the variables actually supplied to the sender.

Before acceleration saturation, the implemented control law is

$$\begin{aligned} u_i = & -K_p s_i \sum_j A_{ij}\big[(p_i - \hat{p}_{ij}) - (r_i - r_j)\big] \\ & - K_v s_i \sum_j A_{ij}(v_i - \hat{v}_{ij}) - \pi_i K_{pL}[(p_i - \hat{p}^L_{i1}) - r_i] \\ & - \pi_i K_{vL}(v_i - \hat{v}^L_{i1}) + \pi_i \hat{a}^L_{i1}. \end{aligned} \tag{5}$$

Its exact-information part is $u^\star_F = -H_p e - H_v w + \Pi \mathbf{1} a_1$; stale memories add a command residual d^c. With $y = \mathrm{col}(e, hw)$, the exact unsaturated sampled update can be written

$$y_{k+1} = A_h y_k + D_h \,\mathrm{col}(\chi_k, \upsilon_k), \tag{6}$$

where χ_k is the analytical-leader position-step residual and $\upsilon_k = h^2 d^c_k + h^2 \Pi \mathbf{1} a_1(k) - h\mathbf{1}[v_1(k+1) - v_1(k)]$. This form follows from the actual velocity-first integrator; it is not a continuous-time approximation.

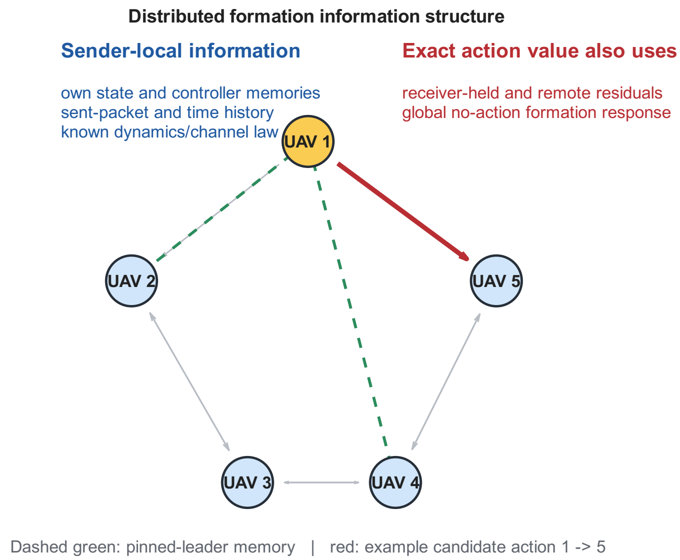

Fig. 1. Evaluated distributed information architecture. The exact value of the highlighted candidate depends on the global no-action response, not only on the sender's own state and estimate of receiver memory.

B. Information available at a sender

At a decision by source j, the strongest deterministic local information used in this paper contains its exact own position and velocity; every ordinary and pinned receiver memory feeding its own controller; its exact unsaturated command; graph, gains, offsets, clock, and channel law; and its own sent-DATA history. The leader additionally knows its reference position, velocity, and acceleration. It does not include state stored at another receiver or remote physical state that has not been communicated. We express the state-dependent part as

$$o_j = C_j \xi + d_j. \tag{7}$$

Known packet histories or action labels change affine constants but do not add hidden-state rows. Figure 1 illustrates the distinction.

IV. Finite-Horizon Value of a Communication Action

A. Generic linear/affine closed loop

Consider a finite-dimensional affine closed loop

$$x_{k+1} = Ax_k + Bu_k + d. \tag{8}$$

For a fixed decision point, collect the no-action performance output over a finite horizon as

$$z = Fx + r. \tag{9}$$

A candidate communication action a induces a fixed response g_a in the same stacked output. With symmetric $Q \geq 0$, define benefit as no-action cost minus action cost:

$$\begin{aligned} q_a(x) &= \|Fx + r\|^2_Q - \|Fx + r + g_a\|^2_Q \\ &= \ell_a^\top x + c_a, \end{aligned} \tag{10}$$

where

$$\ell_a = -2F^\top Q g_a, \qquad c_a = -2r^\top Q g_a - g_a^\top Q g_a. \tag{11}$$

TABLE I
CLOSEST RESEARCH THREADS AND THE DISTINCTION OF THIS PAPER.

Thread	Established capability	Remaining question	Position of this paper
Event-triggered and lossy formation	Stability/performance triggers under distributed state error, loss, or delay	Is exact marginal action value available to the transmitter?	No new trigger; a valid bound-trigger counterexample motivates the information question.
AoI / goal-oriented freshness	Timeliness and incorrectness metrics	Does freshness determine the closed-loop action cross term?	AoI is not treated as a sufficient control-value statistic.
VoI scheduling	Marginal control benefit and threshold scheduling under specified information	Is the value statistic itself identifiable under a distributed sender map?	Studies this precondition; makes no optimal-scheduler claim.
Decentralized information structures	Formal dependence of decisions on distributed histories	Which exact functional is missing for communication decisions?	Instantiates the information question with finite-horizon communication-action value.
Functional and sample-based observability	Row-space/modal/structural tests, sampling conditions, observer design, and target-sensor selection	What irreducible value and decision loss follows when a communication-action functional is hidden?	Uses established tests; adds action-specific minimax regret, simultaneous-action dimension, acquisition separation, and formation closure.

Positive normalizations, including a delivery probability, do not change identifiability or sign. If the response depends on an unknown state, that state must be augmented or the response must be conditioned on a hypothesis; otherwise (10) does not apply. This quantity is a conditional, fixed-response, finite-horizon marginal output-cost benefit. It freezes the decision point, no-action baseline, controller, horizon, and candidate response; it is not a Bellman value, state-action Q-function, or general stochastic VoI that reoptimizes future controls, schedules, or channel outcomes.

B. Formation action response

For the formation case, the augmented state contains exactly the variables needed to construct the frozen no-action response: follower formation error and scaled relative velocity, analytical-leader state, eight ordinary receiver memories, and two pinned-leader memories. For one Cartesian axis,

$$\begin{aligned}\xi^a = \operatorname{col}\big(&e^a, hw^a, p_L^a, hv_L^a, h^2 a_L^a,\\ &\{\hat{p}_{ij}^a, h\hat{v}_{ij}^a\},\\ &\{\hat{p}_{iL}^a, h\hat{v}_{iL}^a, h^2\hat{a}_{iL}^a\}\big).\end{aligned} \tag{12}$$

The one-axis dimension is 33 and $\xi = \operatorname{col}(\xi^x, \xi^y, \xi^z) \in \mathbb{R}^{99}$.

Recursion of (6) over H samples yields implementation-derived matrices, not fitted matrices,

$$\operatorname{vec}(Z_k) = F_H \xi_k + r_H. \tag{13}$$

For link $j \to i$, receiver-memory hypothesis s, successful-delivery probability p_s, $m = N-1$, and fixed output response $g_{ij,s}$,

$$q_{ij,s} = -\frac{2p_s}{m} g_{ij,s}^\top (F_H \xi_k + r_H) - \frac{p_s}{m}\|g_{ij,s}\|_2^2. \tag{14}$$

Here Z_k stacks H sampled follower-position errors. Consequently $q_{ij,s}$ is a per-follower sum of squared sampled position-error differences, measured in m^2; it is not an RMSE or time-integrated mission cost. For a receiver-memory belief $b(s)$, the exact conditional expression is $\bar{q}_{ij} = \sum_s b(s) q_{ij,s}$. The coefficient and constant are averaged across hypotheses. We do not evaluate the nonlinear quadratic at an expected receiver state. When ξ_k contains components hidden from the sender, this receiver-memory marginalization is not $\mathbb{E}[q \mid \mathcal{I}_j]$ unless an additional joint state prior is specified.

V. Information-Structure Identifiability

A. Static current-value reconstructibility

The following classical row-space fact is specialized to the conditional fixed-response communication benefit. It is used as a prerequisite, not claimed as a new functional-observability theorem.

Theorem 1 (Static fixed-response action-value identifiability)**.** *Let $q_a(x) = \ell_a^\top x + c_a$ and $o = Cx + d_o$. On $\mathbb{R}^n$, or locally on a feasible set containing an open relative neighborhood along* $\ker C$, *the exact action value is a function of o if and only if*

$$\ell_a \in \operatorname{row}(C). \tag{15}$$

Proof. If $\ell_a = C^\top \alpha$, then $q_a = \alpha^\top(o - d_o) + c_a$. Conversely, any $v \in \ker C$ leaves o unchanged. Identifiability requires $q_a(x+v) - q_a(x) = \ell_a^\top v = 0$ for every such v. Hence $\ell_a \in (\ker C)^\perp = \operatorname{row}(C)$. □

The normalized diagnostic used later is

$$\delta_C(\ell) = \frac{\|(I - C^\dagger C)\ell\|_2}{\max(\|\ell\|_2, \epsilon)}. \tag{16}$$

A positive numerical residual applies the theorem to a given matrix; it is not itself the theorem.

B. Finite local history

Suppose that during a declared interval the augmented dynamics and measurements are

$$x_{t+1} = A_0 x_t + a_0, \qquad o_t = Cx_t + d. \tag{17}$$

Define

$$\mathcal{O}_L = \operatorname{col}(C, CA_0, \dots, CA_0^L). \tag{18}$$

Theorem 2 (Finite-history action-value identifiability)**.** *At time L, an action value with coefficient ℓ_a is identifiable from $o_0, \dots, o_L$ if and only if*

$$(A_0^L)^\top \ell_a \in \operatorname{row}(\mathcal{O}_L). \tag{19}$$

After the observability row space has stabilized, later samples add no observation direction. They cannot recover a decision-time value whenever its propagated coefficient remains outside that stabilized row space.

Proof. The affine solution gives $x_L = A_0^L x_0 + \sum_{t=0}^{L-1} A_0^{L-1-t} a_0$. Thus the coefficient of the time-L value in x_0 is $(A_0^L)^\top \ell_a$, while the stacked history's state-dependent part is $O_L x_0$. Theorem 1 gives (19). Cayley–Hamilton implies that the row span generated by $C, CA_0, \dots$ stabilizes in finite dimension. □

This is the corresponding sampled functional-observability specialization, not a new result based merely on sampling. The last sentence is conditional on unchanged maps and a coefficient that retains its hidden component. A hidden mode annihilated by the dynamics can cease to affect a later value; that is loss of dependence, not recovery of information. A delivered packet, new sensor, or shared statistic changes the information structure.

C. Irreducible value and decision uncertainty

Exact magnitude is stronger than the sign needed by a transmit-if-beneficial rule. Let $\mathcal{X}$ be the declared operating set,

$$\mathcal{X}(o) = \{x \in \mathcal{X} : Cx + d_o = o\}, \tag{20}$$

and assume its image under q is a nonempty closed finite interval

$$Q(o) = [q_-(o), q_+(o)]. \tag{21}$$

Define the midpoint and value-information radius

$$q_c(o) = \frac{q_+(o) + q_-(o)}{2}, \qquad \Delta_q(o) = \frac{q_+(o) - q_-(o)}{2}. \tag{22}$$

Theorem 3 (Irreducible value and communication-decision uncertainty)**.** *At a fixed information state o:*

1) *the minimax error of any sender-local point estimate is*

$$\inf_{\widehat{q}(o)} \sup_{x \in \mathcal{X}(o)} |\widehat{q}(o) - q(x)| = \Delta_q(o), \tag{23}$$

and the midpoint $\widehat{q} = q_c$ is optimal;

2) *if $q_- < 0 < q_+$, no deterministic map of o recovers the exact value sign for every compatible state;*
3) *for a binary decision whose exact-information rule transmits iff $q > 0$, a sender-local randomized rule transmitting with probability p has local endpoint regrets*

$$R_+(p) = (1-p)q_+, \qquad R_-(p) = p(-q_-). \tag{24}$$

Its minimax probability and unavoidable regret are

$$p^* = \frac{q_+}{q_+ + (-q_-)}, \qquad R^*_{\text{rand}} = \frac{q_+(-q_-)}{q_+ + (-q_-)} > 0. \tag{25}$$

Restricting to deterministic rules gives

$$R^*_{\text{det}} = \min\{q_+, -q_-\} > 0. \tag{26}$$

Proof. For any scalar estimate y, $q_+ - q_- \le |q_+ - y| + |y - q_-|$, so its worst endpoint error is at least $(q_+ - q_-)/2$; the midpoint attains that bound throughout the interval. If the interval straddles zero, the same observation is compatible with opposite exact signs, proving the deterministic statement. For the randomized action, regret over positive values is maximized at q_+ and decreases with p; regret over negative values is maximized at q_- and increases with p. Equating $(1-p)q_+ = p(-q_-)$ yields (25). The deterministic choices $p = 0$ and $p = 1$ incur worst regrets q_+ and $-q_-$, respectively, proving (26). □

The strict-straddling formulas above have transparent boundary cases. If $q_- = 0 < q_+$, choosing $p = 1$ gives zero minimax regret; if $q_- < 0 = q_+$, choosing $p = 0$ does so; and if both endpoints are zero every choice has zero regret. A non-straddling interval admits a deterministic common-sign decision with zero regret. For the symmetric interval $[-a, a]$, $p^* = 1/2$, $R^*_{\text{rand}} = a/2$, and $R^*_{\text{det}} = a$. If transmission instead requires a net benefit above threshold τ, apply the same theorem to $v = q - \tau$ and the shifted interval $[q_- - \tau, q_+ - \tau]$. The formation witnesses below use $\tau = 0$.

For $q(x) = a^\top x + b$, let $P_C = C^\dagger C$, $a_\perp = (I - P_C)a$, and consider the Euclidean local fiber $x = x_0 + v$, $v \in \ker C$, $\|v\|_2 \le \rho$. Its exact interval is

$$\begin{aligned} Q(o) &= [q(x_0) - \rho\|a_\perp\|_2, \\ &\qquad q(x_0) + \rho\|a_\perp\|_2], \\ \Delta_q(o) &= \rho\|a_\perp\|_2. \end{aligned} \tag{27}$$

The endpoints follow by choosing v parallel or antiparallel to $a_\perp$. Figure 5 shows the geometry.

Theorem 3 is a local minimax statement over $\mathcal{X}(o)$, not a universal stochastic-control lower bound. It does not rule out a Bayesian conditional expectation under a specified prior, nor does it say that every information fiber crosses zero.

VI. Minimum Supplementary Information and Coalition Structure

A. One or multiple candidate actions

Stack p action-value coefficients as rows of $L \in \mathbb{R}^{p \times n}$ and define

$$L_\perp = L(I - C^\dagger C). \tag{28}$$

Theorem 4 (Minimum instantaneous linear-statistic dimension)**.** *Under an instantaneous, noiseless, arbitrary-precision real-valued linear supplementary map Hx, the minimum number of scalar rows required to identify all p exact action values is*

$$r_\star = \operatorname{rank}(L_\perp). \tag{29}$$

Proof. If all values are identifiable from $\operatorname{col}(C, H)x$, the projections of the rows of H onto $\ker C$ must span every row of $L_\perp$. Therefore H has at least $\operatorname{rank}(L_\perp)$ independent scalar rows. Conversely, choose the rows of H as any row basis of $L_\perp$. Each row of L is then the sum of a component in $\operatorname{row}(C)$ and one in $\operatorname{row}(H)$, satisfying Theorem 1 for all p values. □

For a single nonidentifiable value, one additional scalar $\theta = \ell_\perp^\top x$ is sufficient. This is statistic dimension, not packet count. The coordinates of $\ell_\perp$ may be distributed across many agents.

Corollary 1 (Instantaneous linear supplementary statistics)**.** *Under the same noiseless arbitrary-precision linear model, if the decision maker can receive r independent real-valued scalar statistics before deciding, exact identification of all p values requires $r \ge r_\star$. Equality is algebraically achievable if those statistics span* $\operatorname{row}(L_\perp)$.

This is a linear-statistic-dimension statement, not a data-rate theorem. Scalar dimension, packet count, bits, latency, functional-observer order, and the distributed cost of synthesizing each scalar are distinct quantities.

B. Information ownership

Let agent i's strongest map be C_i. For coalition $\mathcal{S}$, define $C_{\mathcal{S}} = \mathrm{col}_{i\in\mathcal{S}}\, C_i$.

Corollary 2 (Coalition identifiability)**.** *Coalition $\mathcal{S}$ identifies a fixed action value if and only if*

$$\ell \in \mathrm{row}(C_{\mathcal{S}}). \tag{30}$$

A minimum owning coalition solves

$$\min_{\mathcal{S}} |\mathcal{S}| \quad \textit{subject to} \quad \ell \in \mathrm{row}(C_{\mathcal{S}}). \tag{31}$$

Corollary 2 is closely related to functional sensor selection. We use exhaustive enumeration only for the N=5 case and do not claim a new generic combinatorial algorithm. Even when a coalition owns the required rows, a causal aggregation protocol must still communicate them at a nonzero frequency, delay, and reliability cost.

VII. Communication Cost of Acquiring Value Information

Let $C_P(E)$ be the cost of the frozen periodic Pareto frontier interpolated at formation error E. Suppose a policy informed by supplementary statistics uses DATA cost D and information streams t with rate R_t and normalized price η_t. A necessary accounting condition for performance-matched Pareto benefit is

$$D + \sum_t \eta_t R_t < C_P(E). \tag{32}$$

For a single feedback stream with rate A,

$$\eta^\star = \frac{C_P(E) - D}{A} \tag{33}$$

is its break-even price. Negative $\eta^\star$ means DATA alone is already more costly than periodic at matched error. At feedback price η_f, the affordable feedback fraction is $f^\star = \eta^\star/\eta_f$, clipped to $[0, 1]$ for operational interpretation.

Equation (32) is not an optimality theorem. It prevents a common category error: a one-dimensional sufficient statistic is not free. Under a packet-frequency metric, piggybacking adds no packet only if an existing packet is sent anyway; bytes, airtime, collision risk, and joules remain outside that metric.

VIII. Formation-Control Case Study

A. Implementation-derived matrices and cross-instance validation

The frozen case uses N=5, $h = 0.02$ s, four followers, eight ordinary controller links, two pinned-leader streams, H=25, and D=4 samples. Its augmented state has 33 entries per axis and 99 in 3-D. The matrices F_H, action responses, sender maps, and held-memory dynamics are assembled directly from the implemented graph, gains, controller, and integrator. For each controller-relevant payload, a structural command correction $[1, 0, 0]^\top$ and fixed receiver-memory hypothesis define g. The maximum discrepancy between the affine representation and the centralized calculation is 4.34×10^{-16}.

To test whether the information-structure observation is peculiar to that instance, we froze a new Cartesian registry before viewing its results: $N \in \{5, 7, 9\}$, three deterministic connected graphs, even/odd follower pinning, $H \in \{15, 25\}$, and $D \in \{2, 4\}$. All 72 configurations satisfy the same symmetric grounded Schur scope. Fixed-payload histories use the complete finite-dimensional horizon $n_{\mathrm{free}} - 1$. Table II summarizes all 1320 action rows.

At 10^{-10}, compressed-null and independent direct-row-space calculations agree for 1320/1320 rows; the normalized hidden residual spans 3.03×10^{-6}–0.99999. Across the tolerance grid, raw rank is stable for only 608/1320 rows and raw rank$([C; \ell^\top])$ adds one in 1222 rows at 10^{-6} and 614 at 10^{-14}. An independent reconstruction matches all 6600 stored decisions, while normalized-basis augmentation gives 1320/1320 at every cutoff. Selected 80-digit checks, including each cell's minimum, remain positive. The claim is therefore scale-controlled functional membership in a finite registry, not invariant raw-rank bookkeeping or an arbitrary-graph/gain theorem.

The structural correction fixes g and must not be confused with a trajectory-realized scheduling action. Zero response has zero value, and special nonzero responses can cancel a hidden projection. The next subsection therefore audits actual state-generated corrections separately.

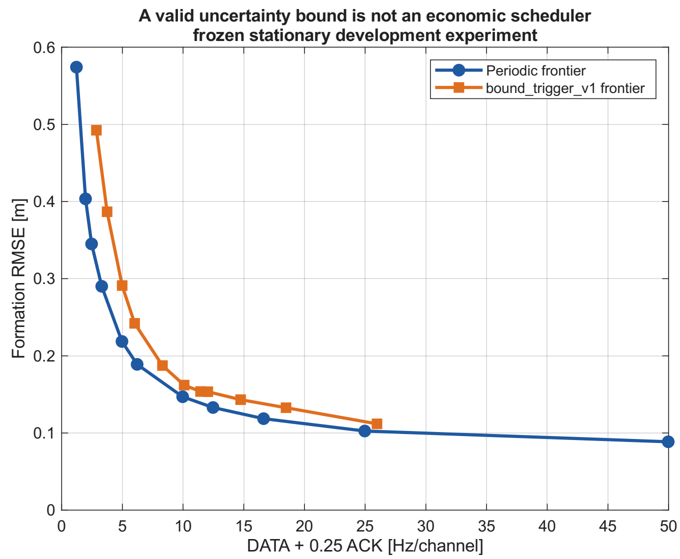

Fig. 2. Frozen stationary frontiers from five paired development seeds. The analytically valid bound-based mechanism is dominated by the periodic frontier over their shared domain. The comparison is descriptive, not a population claim.

B. Reachable value-sign witnesses

Affine nonidentifiability alone could be dismissed as an inadmissible state perturbation. We therefore audit every controller-relevant action in the frozen N=5 catalog. Receiver memories are initialized consistently and held without a DATA update until the decision. A deterministic source excitation creates an actual 0.01 m/s^2 x-axis controller correction. For follower senders, the larger of the source's position and scaled-velocity transfers is used; leader actions use constant velocity. We then project the exact fixed-response value gradient into the

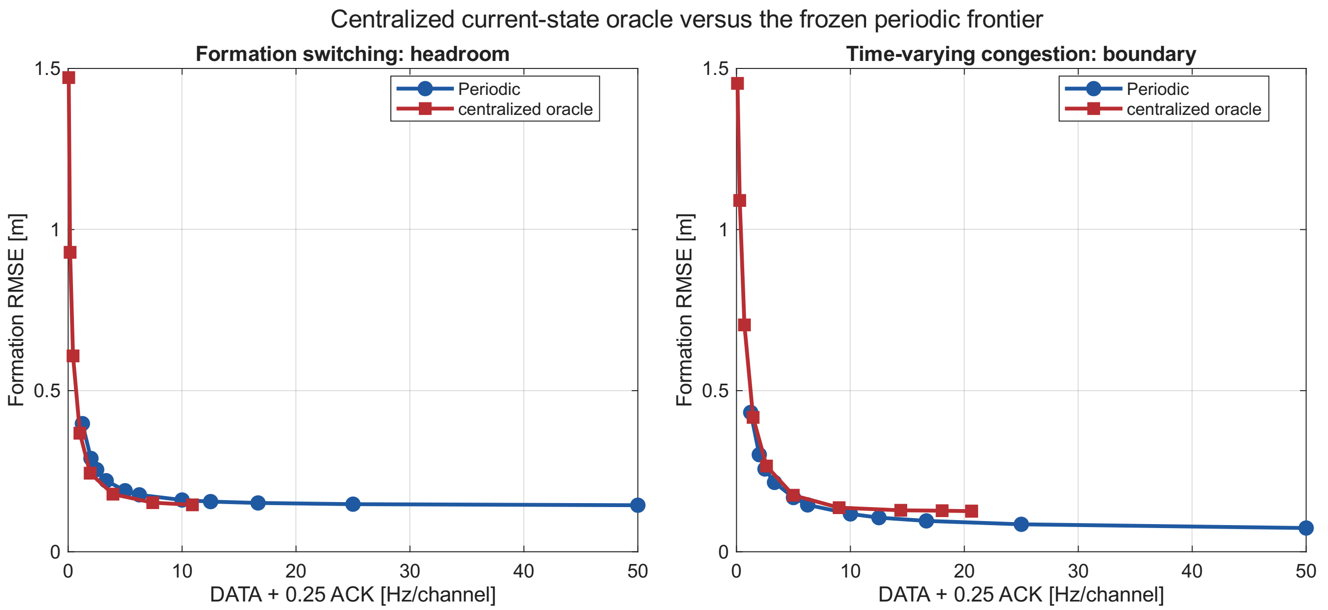

Fig. 3. Frozen five-seed development diagnostic with centralized current plant and receiver information. The conditional fixed-response value creates Pareto headroom in S2 but not in the S4 congestion boundary. The oracle is privileged and nondeployable, and the comparison is non-inferential.

TABLE II

Cross-instance structural audit. "Rows" aggregates two pinning patterns, two H values, and two D values; every row is nonidentifiable, stable in membership over relative tolerances 10^{-6}–10^{-14}. After the retained row space is orthonormalized and the value row is normalized, every value row adds one dimension. "Raw stable" counts rows whose unnormalized information-map rank itself does not change over that grid.

Graph	N	Rows/non-ID	Raw stable	Leader (p_1, r_1^*)	r_1^*/p_1	Minimum coalition pattern
geometric	5	48/48	48	(6,4)	0.667	leader: 5
ring2	5	80/80	56	(4,3)	0.750	every sender: 5
sparse4	5	144/144	96	(6,4)	0.667	every sender: 5
geometric	7	104/104	80	(7,5)	0.714	leader: 7; followers: 5
ring2	7	120/120	80	(5,4)	0.800	every sender: 7
sparse4	7	216/216	56	(7,5)	0.714	every sender: 7
geometric	9	160/160	80	(8,6)	0.750	leader: 9; followers: 8
ring2	9	160/160	48	(6,5)	0.833	every sender: 9
sparse4	9	288/288	64	(8,6)	0.750	every sender: 9

consistent-initialization subspace that simultaneously preserves the complete sender history, target receiver memory, and actual action response. Symmetric perturbations about the exact affine zero crossing are replayed through the implemented controller and integrator.

All ten actions yield admissible opposite-sign pairs, including six follower- sender actions not covered by the earlier representative construction. Independent replay from persisted coordinates recomputes the physical controller and centralized trajectories rather than reusing the affine sign expression. Correction, payload, branch, and response match within every pair. Maximum history, dynamic, and affine-versus-oracle residuals are 7.98×10^{-20}, 4.17×10^{-16}, and 4.07×10^{-20}; response mismatch is zero. The 10^{-4} coordinate half-separation yields a smallest endpoint 8.52×10^{-8}, over 8.5×10^{4} sign tolerances, and minimum saturation margin 0.952 m/s^2. Values in Table III are in m^2 under the sampled per-follower normalization following (14).

Because the endpoints are centered symmetrically, Theorem 3 gives $p^* = 1/2$, randomized minimax regret $\Delta_q/2$, and deterministic minimax regret Δ_q for every row. These are local one-decision losses, not expected mission-level bounds. Practical scale is heterogeneous: the normalized radius exceeds 0.15 for both pin and both leader ordinary actions, but is below 3.5×10^{-4} for two follower actions. Seven of ten actions cross zero somewhere on the selected 32-state grid; the formal reachable witness exists for all ten. The maximum is 0.232, or 0.145 and 0.175 under RMS and IQR/1.349 scaling; the two smallest remain order 10^{-4}. Figure 6 shows this heterogeneity.

C. Statistic dimension versus coalition size

For every fixed payload, $\mathrm{rank}(L_\perp) = 1$: one linear scalar closes its individual algebraic deficit. However, no receiver alone and no sender–receiver pair identifies that scalar. Exhaustive enumeration of all subsets of the five implemented agent maps finds that every sender needs four additional agents; the only minimum coalition contains all five UAVs. This is conditioned on the evaluated graph, H=25, the selected global stacked formation-error output with identity weighting, and the declared current information maps. A localized or decomposable objective can yield a smaller coalition. It is not a claim that all formations require global information.

The fixed-action statement underuses Theorem 4. We therefore stack every controller-relevant current action of each

TABLE III

N=5 ACTUAL-ACTION REACHABLE VALUE-SIGN WITNESSES. THE NORMALIZED RADIUS DIVIDES Δ_q BY THE MEDIAN ABSOLUTE EXACT VALUE ON A DETERMINISTIC 32-STATE LOCAL GRID. "CROSS" IS THE FRACTION OF THOSE STATES WHOSE RADIUS- 10^{-4} COMPATIBLE INTERVAL CROSSES ZERO; IT IS DESCRIPTIVE, NOT A POPULATION PROBABILITY.

Action	q_-	q_+	Δ_q	Normalized radius	Cross
ordinary $1 \to 2$	-3.475×10^{-6}	3.475×10^{-6}	3.475×10^{-6}	0.1590	0.1875
ordinary $1 \to 5$	-5.344×10^{-6}	5.344×10^{-6}	5.344×10^{-6}	0.1953	0.0625
ordinary $2 \to 3$	-5.392×10^{-6}	5.392×10^{-6}	5.392×10^{-6}	0.01010	0.09375
ordinary $3 \to 2$	-8.519×10^{-8}	8.519×10^{-8}	8.519×10^{-8}	0.0002666	0
ordinary $3 \to 4$	-3.198×10^{-6}	3.198×10^{-6}	3.198×10^{-6}	0.009173	0.0625
ordinary $4 \to 3$	-5.223×10^{-6}	5.223×10^{-6}	5.223×10^{-6}	0.01080	0
ordinary $4 \to 5$	-1.333×10^{-7}	1.333×10^{-7}	1.333×10^{-7}	0.0003459	0
ordinary $5 \to 4$	-3.396×10^{-6}	3.396×10^{-6}	3.396×10^{-6}	0.01099	0.0625
pin $1 \to 2$	-3.449×10^{-6}	3.449×10^{-6}	3.449×10^{-6}	0.2315	0.2500
pin $1 \to 4$	-5.002×10^{-6}	5.002×10^{-6}	5.002×10^{-6}	0.2106	0.03125

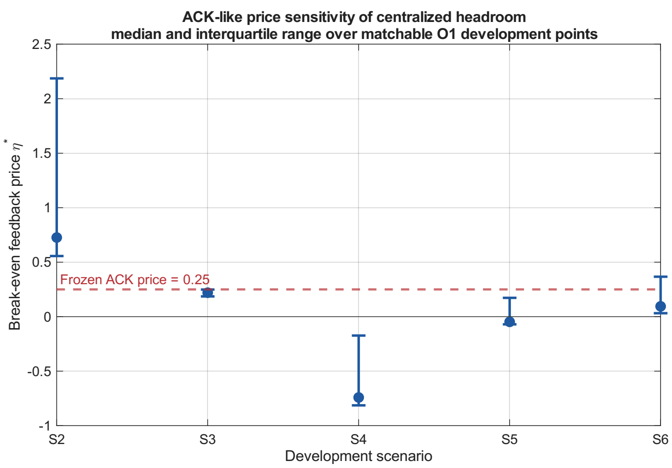

Fig. 4. ACK-like packet-frequency cost sensitivity over frozen, performance-matchable centralized-oracle development points (median and IQR). The dashed line is the preregistered price 0.25. This does not measure the cost of acquiring the all-agent missing statistic.

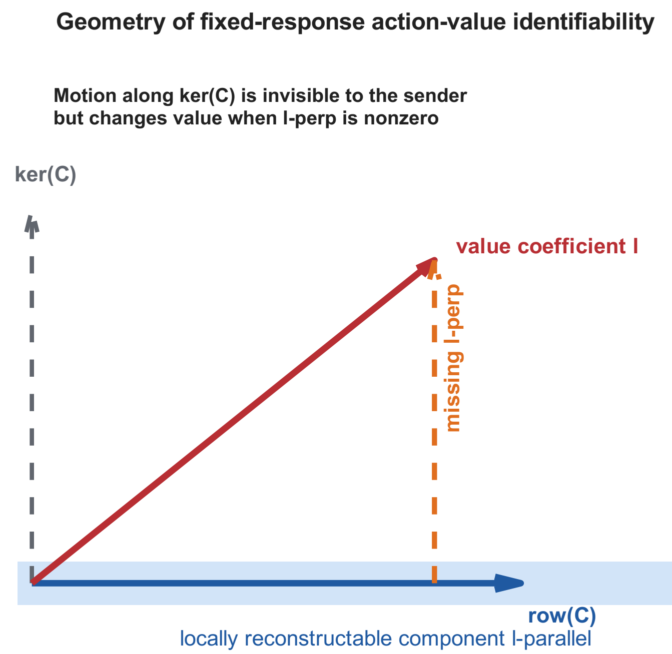

Fig. 5. The available row space determines the reconstructable component of the value coefficient. Motion along the information null space is invisible to the sender but changes value when the missing projection is nonzero.

actual sender in L_j, using a common full 99-dimensional state and the structural unit command response. Table IV reports the simultaneous missing rank. Sender 1 has four payloads but $r_1^* = 3$: its ordinary and pinned-leader unit actions into follower 2 induce the same output-response direction, so one hidden direction is shared. Senders 3 and 4 each require two independent directions for two actions; the single-action senders require one. Thus compression exists for the leader but is not universal. The nonzero singular values shown in the table are those of $L_j(I - C_j^\dagger C_j)$.

Although $r_1^* = 3 < 4$, the minimum coalition for its entire missing subspace still contains all five agents. The same coalition size holds for every other sender's full action set. Compression of value directions therefore does not imply local ownership or three packets: Corollary 1 counts independent real-valued statistics only if their operands can be assembled before the decision.

The cross-instance audit qualifies the ownership result. Leader compression recurs in every registry cell: r_1^*/p_1 ranges from 2/3 to 5/6, while every tested follower has $r_j^* = p_j$. Ring2 and sparse4 require the complete N-agent coalition for every sender. On the geometric graph, however, follower-sender value sets require five of seven agents and eight of nine agents, while the leader still requires all agents. Coalition ownership is therefore topology and action-set dependent; the general claim is the row-space condition, not an all-agent rule. Across 448 sender–cell rows, unique minima 5, 7, 8, and 9 occur 120, 120, 56, and 152 times; these counts do not imply packet or acquisition cost.

D. Why the information result matters operationally

Figure 2 retains the mechanism that failed before this theory was formulated. Its trigger uses a valid staleness-to-control uncertainty bound, yet the complete periodic frontier is better throughout the shared development domain. Across a 101-point budget-matched grid, trigger-minus- periodic RMSE averages +0.03023 m and no point favors the trigger. This is a counterexample to the inference that a safe uncertainty bound is automatically a useful scheduler; it is not a population comparison.

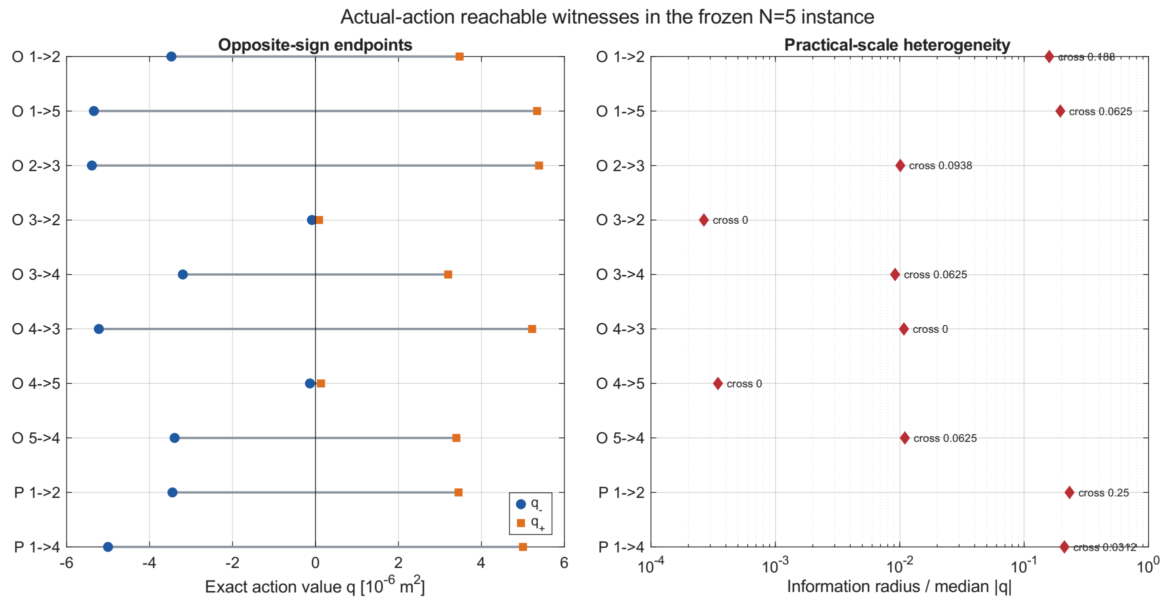

Fig. 6. Implementation-replayed N=5 actual-action witnesses. Left: exact opposite-sign endpoints for all ten controller actions. Right: local information radius normalized by typical exact action-value scale; labels give the deterministic-grid zero-crossing fraction. Small normalized radius for some actions is retained rather than interpreted as uniform practical impact.

TABLE IV

Sender-wise simultaneous-action missing dimension and ownership.

Sender j	p_j	r_j^*	r_j^*/p_j	Retained singular values of $L_j^{\perp}$	Minimum coalition	Interpretation
1	4	3	0.75	26.9758; 20.5237; 1.83164	$\{1, 2, 3, 4, 5\}$	one shared direction
2	1	1	1.00	49.7427	$\{1, 2, 3, 4, 5\}$	single action
3	2	2	1.00	55.0690; 16.5170	$\{1, 2, 3, 4, 5\}$	independent directions
4	2	2	1.00	61.7396; 5.34031	$\{1, 2, 3, 4, 5\}$	independent directions
5	1	1	1.00	45.7777	$\{1, 2, 3, 4, 5\}$	single action

TABLE V

Frozen ACK-like packet-frequency economics. Statistics cover only centralized-oracle development points within the measured periodic performance domain and do not price a protocol for acquiring the all-agent missing value statistic.

Item	Statistic	Result	Interpretation
S2 formation switching	eligible 4/14; median $\eta^{\star}$	0.726 (IQR 0.556–2.186)	Full feedback at price 0.25 is affordable for every eligible point.
S3 burst loss	eligible 11/14; median $\eta^{\star}$	0.220 (0.186–0.249)	Median point cannot afford full feedback.
S4 congestion	eligible 11/14; median $\eta^{\star}$	-0.741 ($[-0.814, -0.173]$)	DATA alone is uneconomic throughout the eligible set.
S5 topology perturbation	eligible 11/14; median $\eta^{\star}$	-0.047 (-0.070–0.173)	Headroom is sparse and not robust to feedback cost.
S6 dynamic excitation	eligible 11/14; median $\eta^{\star}$	0.095 (0.032–0.367)	Median affordable feedback fraction is 37.8%.

To test whether valuable scheduling opportunities existed at all, a centralized online oracle used exact current plant and receiver memories but no future channel outcomes, disturbances, formation changes, or trajectories. It ranked actions only by (14). Figure 3 shows a positive and a negative boundary. In S2 formation switching, oracle minus periodic mean matched RMSE is -0.02381 m and mean matched cost is -1.7088 Hz/channel; the oracle is favored over the complete shared domains. In S4 time-varying congestion, the differences reverse to +0.02595 m and +0.7135 Hz/channel, with zero shared-domain fraction favoring the oracle. The oracle is privileged and is not a proposed algorithm.

Finally, Fig. 4 applies (33) to an ACK-like receiver-information stream. At the frozen ACK price 0.25, S2 can fund one such packet per accepted delivery at all four matchable points, while the median S6 point can fund it on only 37.8% of eligible deliveries. S4 has negative headroom even before feedback. Table V reports all five scenarios. This is a necessary cost-sensitivity screen, not the measured cost of causally aggregating the all-agent statistic. It explains why algebraic sufficiency and deployable economic value must remain distinct.

IX. Discussion and Limitations

A. What the result changes

Theorems 1–4 suggest a three-stage audit for control-aware communication. First derive the conditional fixed-response action benefit under the closed-loop model. Second test whether that functional lies in the acting agent's information row space, including the complete permitted history. Third, if information is missing, distinguish the algebraic dimension of the missing statistic from the coalition and communication resources needed to construct it. Scheduler design before these checks risks optimizing a surrogate that either lacks the decisive cross term or costs more to reveal than its scheduling benefit.

The result also clarifies why receiver-memory estimation alone can be insufficient. Such an estimator can correctly represent which packet the receiver may hold. The finite-horizon action value additionally projects the action response onto the no-action closed-loop response, which includes formation components outside the sender's information. Better calibration of the receiver-memory marginal cannot create measurements of those missing components.

B. Boundaries

The analytical formation case is the unsaturated sampled double-integrator subsystem with a fixed graph during each decision history. It is not a theorem for nonlinear 6-DOF UAV dynamics, saturation, switching topology within a history, or an interference-limited MAC. The 72-cell audit establishes recurrence over a deterministic finite registry, not arbitrary N, gains, or graphs. Its coalition results are explicitly heterogeneous. Tolerance sweeps and selected 80-digit projections harden the numerical membership conclusions but do not replace a symbolic arbitrary-parameter proof. The minimum residual is only 3.03 times the loosest declared tolerance, although its 80-digit conditional-basis check agrees to 10^{-16}; this is a numerical margin, not a structural lower bound.

The value-radius and regret results are local to an information-compatible operating set. A fully specified stochastic prior and nonlinear Bayesian estimator may produce a useful conditional expected value; that is a different information model and is not ruled out, and this paper does not compare against the best policy in that class. Randomization reduces worst-case binary regret for the ten centered N=5 witnesses but cannot remove it or make the hidden exact sign observable. Their normalized ambiguity varies by nearly three orders of magnitude, so existence is not uniform engineering impact.

The frontier evidence uses five paired development seeds and is presented as mechanism falsification and counterexample evidence. No held-out population superiority or statistical significance claim is made. The communication axis counts DATA + 0.25 ACK packets per second per channel. It omits payload length, shared-medium collisions, airtime, and energy, so feedback-economic numbers are conditional on this accounting.

Functional, sample-based, and structural functional observability, functional observer design, and target-sensor selection are established fields. Our exact row-space, finite-history, and coalition conditions must be read as action-value specializations, not as new generic observability or placement theory. The intended contribution is the control-communication synthesis: the exact action functional, irreducible local value/decision loss, shared missing dimensions across simultaneous actions, acquisition cost, and implementation-faithful reachable closure.

X. Conclusion

Control-aware communication is not locally decidable merely because a sender can estimate the receiver's memory. For a fixed communication response in a linear sampled closed loop, quadratic finite-horizon benefit is an affine state functional. Established functional-observability geometry then gives an exact identifiability test; its hidden projection quantifies local minimax value error and strictly positive communication-decision regret when the compatible interval straddles zero. For several action values, the rank of their joint hidden projection gives the minimum instantaneous linear-statistic dimension. The acquisition problem is separate: even a compressed deficit may be distributed across a large coalition and may consume the Pareto headroom it is meant to exploit.

Across the 72-cell formation registry, all 1320 structural fixed-response functionals violate sender-history identifiability under the declared maps. In the evaluated five-UAV topology, all ten actual controller actions also admit reachable, unsaturated history pairs requiring opposite exact decisions while remaining sender-indistinguishable with identical responses. Their normalized information radii range from 2.67×10^{-4} to 0.232, so the existence result is not asserted to have uniform mission-level impact. Four leader actions share three missing directions in N=5; leader compression recurs in the finite registry, whereas tested follower action sets have none. Coalition size varies with topology. These results motivate information-structure and acquisition-cost audits before a value-based scheduler is engineered. Future work may study structured priors, nonlinear functional observability, and causal protocols for acquiring the missing statistic, but none is claimed here.

Appendix A
Detailed Formation Affine Map

For one axis, let $T_y\xi = y$ and write the exact held-current communication input as $T_v\xi + t_v$. Initialize $M_0 = T_y$ and $b_0 = 0$. Then recurse as

$$M_r = A_h M_{r-1} + \begin{bmatrix} I \\ I \end{bmatrix} T_v, \tag{34}$$

$$b_r = A_h b_{r-1} + \begin{bmatrix} I \\ I \end{bmatrix} t_v. \tag{35}$$

With $C_e = [I\ 0]$, stacking $C_e M_r$ and $C_e b_r$ for $r = 1, \ldots, H$ and then for three coordinates yields F_H and r_H in (13). Substitution of a fixed receiver memory removes its coordinates from the free state and moves them into the constant. The action response is generated by the same A_h, delay D, gains, and target row used by the centralized diagnostic.

Appendix B
Euclidean-Fiber Interval Proof

For $v \in \ker C$, $a^\top v = a_\perp^\top v$. Cauchy–Schwarz gives $|a_\perp^\top v| \leq \rho \|a_\perp\|_2$. If $a_\perp \neq 0$, equality is attained at $v = \pm\rho a_\perp / \|a_\perp\|_2$, which lies in $\ker C$. If $a_\perp = 0$, the interval collapses to one point. This proves (27).

Appendix C
Reproducibility Scope

The base theorem validation entry point is `experiments/tcns_information_limits_validation.m`; the R1 depth audit is `experiments/tcns_r1_theory_depth_validation.m`, with unit test `tests/test_tcns_r1_theory_depth.m`. Figure generation is `paper/tcns/manuscript/make_gm_figures.m`. The authoritative base theorem-validation run, generated from source commit `15177a8`, is `results/tcns_information_limits_validation/2026-09-08_095209`. The authoritative R1 rank/regret/multi-action audit, generated from source commit `05fa200`, is `results/tcns_r1_theory_depth_validation/2026-09-08_103325`. The authoritative R2 cross-instance and all-action witness audit, generated from implementation commit `a2cec3b`, is `results/tcns_r2_generalization_validation/2026-09-08_161130`. The independent R2.5 audit is `results/tcns_r2_5_adversarial_validation/2026-09-08_175427`. Registry, witness-scenario, and reference-direction seeds are 27022001, 27020001, and 27022001 plus action index; no stochastic draw enters the witness construction. Frozen negative-trigger, centralized-oracle, feedback-accounting, and feedback-economic result directories are named in the repository's claim–evidence ledger. Each result records configuration, seed, source commit, MATLAB release, machine-readable tables, and a console log. No held-out seed or new policy experiment is used in this manuscript.

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